

Knowledge Distillation and Weight Pruning for Two-step Compression of ConvNets in Rice Leaf Disease Classification

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Abstract

Rice is a crucial agricultural crop, but it is susceptible to various diseases. Although Convolutional Neural Networks (CNN) excel in classifying plant diseases, their significant computational demands render them impractical for resource-limited devices and real-time applications. This study presents a two-step model compression pipeline that integrates Weight Pruning and Knowledge Distillation, effectively reducing the size and number of parameters in CNNs while maintaining performance. The optimized models facilitate real-time analysis of leaf images, enabling agricultural researchers to enhance crop productivity, minimize losses, and promote sustainable farming practices. In this study, DenseNet169 demonstrates superior performance among the base models, achieving an accuracy of 99.4%. Meanwhile, ResNet50 outperforms other models as both the Student and Pruned Models, recording an accuracy of 99.5% in both scenarios.

CCS Concepts

• **Computing methodologies** → **Neural networks; Computer vision problems.**

Keywords

Convolutional Neural Networks, Weight Pruning, Knowledge Distillation, Rice Leaf Disease Classification, Deep Learning, Image Analysis

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1 Introduction

Rice being an inevitable significant crop, faces a great challenge due to various diseases and infestation. The challenges include a severe reduction in crop yielding, and a drop in quality and overall productivity. The delay in detecting these diseases results in economic losses and food insecurity. Over the past five years, more than 800 million people have experienced food insecurity [13]. According to the International Rice Research Institute, common diseases like bacterial leaf blight can reduce the yield by up to 70% [1] and Brown spot disease can damage rice production by up to 45% [2]. Therefore, timely and accurate detection and analysis of rice leaf diseases are critical for ensuring food security [29]. Traditionally, disease identification relies on professionals, leading to high labor costs, limited monitoring, and potential errors in identification, which complicate disease detection in agriculture [8]. Consequently, achieving automated and precise detection of these diseases, without human intervention, remains a significant challenge in precision agriculture [7].

Deep learning techniques have already demonstrated their advantages in agriculture, particularly in classifying rice leaf diseases. Among these methods, Convolutional Neural Networks have experienced a resurgence due to their ability to extract features without requiring segmented operations while still achieving satisfactory performance. However, CNNs are computationally intensive, necessitating substantial processing power and storage, which can pose challenges in resource-constrained environments like farms or rural areas, where access to advanced computing resources is limited. This is where model compression techniques, such as weight pruning and knowledge distillation, become essential. Weight pruning eliminates redundant or less critical connections within the network, thereby reducing model size and computational requirements without compromising performance. In contrast, knowledge distillation involves transferring knowledge from a larger, more

complex model (teacher) to a smaller, more efficient model (student), allowing for similar accuracy while significantly lowering complexity.

Numerous studies have demonstrated the application of Convolutional Neural Networks (CNNs) for classifying plant leaf diseases, as well as the use of Knowledge Distillation and weight pruning for CNN compression. Advances in deep learning have significantly improved computer technology's capabilities in image classification and object recognition [17, 23, 28]. One study [18] proposed a novel classification model for plant leaves utilizing CNN's feature extraction abilities. By employing data augmentation, they developed a ten-layer CNN model that achieved an accuracy of 88%. Another study [31] applied pruning techniques to reduce the computational complexity of DenseNet, resulting in a pruned model that outperformed the original. Additionally, the use of Knowledge Distillation to enhance the inference time of the CNN model MobileNet demonstrated that pruning techniques can be effectively implemented without sacrificing accuracy [24].

The primary motivation for this work stems from the need to address the limitations of deploying large, complex models in real-world agricultural environments. While state-of-the-art deep learning models excel in accuracy, their high computational and memory demands make them impractical not only for edge devices but also for autonomous farming systems, where real-time processing and efficiency are critical for seamless operation in dynamic environments. By reducing model size and computational complexity without compromising performance, we aim to bridge this gap and create an efficient, scalable solution that supports the broader goals of agricultural automation and research advancements.

This research yields the following contributions, as outlined in this paper

- We introduce a dual-step model compression approach combining knowledge distillation and weight pruning, resulting in a significant compression in size for various CNN models.
- This research offers an adaptation of CNN Models for real-time deployment by tailoring compressed versions of VGG16, ResNet50, and DenseNet169.
- Our proposed methodology demonstrates a significant improvement in resource efficiency while maintaining inference time, advancing real-time disease monitoring in resource-constrained agricultural environments.

We organize the rest of this study as follows: We begin with some of the most recent works on rice leaf disease classification problems in Section II. Following that, Section III illustrates the approach we introduce, a two-step compression pipeline. Section IV shows the experiments in detail. Finally, in Section V, we discuss and demonstrate possible future implications.

2 Related Works

2.1 Plant Leaf Disease and Image Classification using CNN

An ensemble model, DEX, combining XceptionNet, EfficientNetB7, and DenseNet121, achieved 98% accuracy in detecting common rice diseases in Bangladesh. Among other CNN architectures tested, Inceptionv3 and DenseNet101 showed the highest accuracy at 97%,

while SeresNext101 initially performed the lowest at 79%, but improved to 96% with transfer learning [4]. Another study presents a CNN-based model for rice leaf disease recognition that reduces network parameters and achieves 99.78% training accuracy, 97.35% validation accuracy, and 97.82% accuracy on independent datasets with an AUC of 0.99. In binary classification, it achieves recognition rates of 97%, 96%, 96%, 93%, and 95% for Blast, Brownspot, Bacterial Leaf Blight, Sheath Blight, and Tungro, respectively [10].

Additionally, research focuses on automating paddy leaf disease detection for four common diseases in Bangladesh, using CNN models to improve accuracy over traditional methods. The study evaluates VGG-19, Inception-ResNet-V2, ResNet-101, and Xception models, achieving the highest accuracy of 92.68% with Inception-ResNet-V2. Early detection significantly reduces crop losses and provides economic support to farmers [12]. Another study tackled the challenge of rice disease identification by proposing a CNN-based method for identifying 10 common rice diseases using a dataset of 500 images of rice leaves and stems, achieving 95.48% accuracy under 10-fold cross-validation. The approach significantly outperforms conventional machine learning models, demonstrating the effectiveness of deep learning in rice disease diagnosis [19]. Further research applied a transfer learning-based Deep Convolutional Neural Network (DCNN) approach, modified from VGG19, for detecting and classifying six rice leaf disease classes. Using a non-normalized augmented dataset, the model achieves an accuracy of 96.08%, with a precision of 0.9620, recall of 0.9617, specificity of 0.9921, and F1-score of 0.9616. This method outperforms similar models on comparable datasets. [15]

2.2 Weight Pruning and Knowledge Distillation

A study proposes an "Entropy-based pruning" method for CNNs, dynamically adjusting pruning thresholds during training based on weight contributions, improving compression [11]. Applied to LeNet-5 on the MNIST dataset, it achieves a 28.25% higher compression rate without loss of accuracy. Then, another study proposes combining Knowledge Distillation with a pruning filter to compress CNN models, which gets applied to MobileNets. After reducing the model size by 26.08%, accuracy increases from 63.65% without KD to 65.37% with KD, though the model without KD had a 0.1s faster inference time [31]. A study develops a CNN architecture with minimal parameters, using six two-dimensional convolutional layers and dropout to prevent over-fitting. The model, optimized with Adam and a 0.001 learning rate, achieves 98.18% accuracy with enhanced anomaly detection and computational efficiency over 50 epochs [24]. Another paper proposes a lightweight convolutional neural network model named PLeaD-Net for plant leaf disease classification. It achieves a testing accuracy of 98.18% on the PlantVillage dataset, with significantly fewer parameters compared to other deep learning models. [21]

It is evident that, while some methods achieve high accuracy, they are often computationally expensive and not sufficiently optimized for resource-constrained environments, limiting their practical implementation at scale. Additionally, optimization techniques like weight pruning and knowledge distillation are not explored extensively in the rice leaf disease classification domain. These limitations motivate us to develop an effective model compression

approach for CNN. To the best of our knowledge, our proposed methodology is the first to focus on a two-step compression pipeline for ConvNets, combining Knowledge Distillation and Weight Pruning, specifically aimed at achieving resource-efficient rice leaf disease classification.

3 Methodology and Architectures

3.1 Workflow of the Study

Our work focuses on compressing Convolutional Neural Networks (CNNs) for plant leaf disease image classification by employing a two-step compression pipeline that integrates Knowledge Distillation and Weight Pruning. This pipeline involves several critical stages, including selecting suitable teacher models, designing lightweight student models, implementing an appropriate distillation loss function, and applying an adaptive weight pruning strategy to further reduce the student model's complexity. In the following subsections, we delve deeper into each component of our methodology, outlining the algorithms and techniques employed in detail. In our research, we employ the pre-trained (Trained on Imagenet dataset) version of three widely adopted convolutional neural network (CNN) architectures– VGG16, ResNet50, and DenseNet169 as teacher models for knowledge distillation due to their proven performance in visual recognition tasks [6] and suitability for plant disease classification [14].

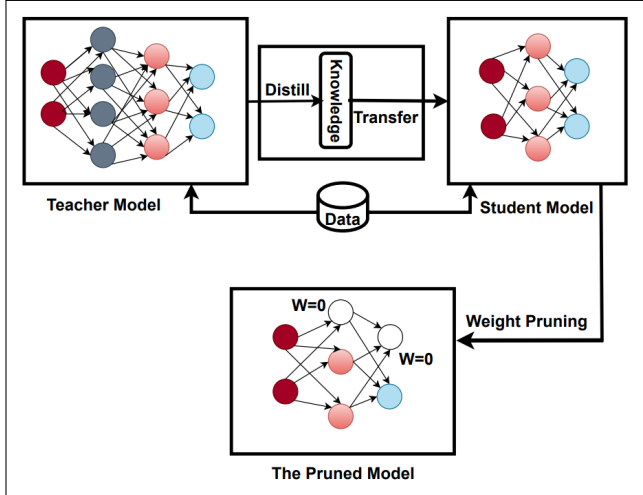


Figure 1: The Two-step compression pipeline

VGG16 is known for its deep architecture with uniform convolutional layers, which effectively capture both low and high-level features [30]. The deep residual learning framework along with the identity skip connection feature of ResNet50 allows it to perform well in fine-grained classification tasks [20]. Whereas DenseNet169 utilizes densely connected layers to enhance feature reuse and gradient flow efficiency making it a good fit for solving classification problems.

3.1.1 Proposed Dual step Compression Pipeline. As illustrated in Figure 1, the process begins with a teacher model, pre-trained on a larger dataset, transferring its learned knowledge to a

more efficient student model via knowledge distillation. In the next step, selective threshold-based weight pruning is applied to specific layers of the student model, enhancing its efficiency without compromising accuracy. The final pruned student model is then evaluated and fine-tuned if necessary to ensure an optimal balance between performance and computational efficiency.

3.2 Knowledge Distillation

3.2.1 The Distillation Loss Function. To create smaller yet efficient student models, we utilized a custom distillation loss function that effectively balances two critical components: the soft loss, which drives the student model to mimic the teacher's output distribution, and the hard loss, which ensures the student's predictions align with the true labels. We apply temperature scaling to both teacher and student logits to smooth the output probability distributions. This allows the student model to capture more informative signals from the teacher model's softened predictions, which is particularly beneficial in learning finer details of the teacher's knowledge [16]. The soft loss is computed using Kullback-Leibler (KL) divergence [5] between the softened outputs of the teacher and student models, ensuring that the student model closely approximates the teacher's behavior. Simultaneously, the hard loss is calculated using the standard cross-entropy loss, aligning the student's predictions with the ground truth labels to retain classification accuracy. Then the two loss components are combined using a tunable weighting parameter, α that controls the relative importance of the soft and hard losses. Adjusting this parameter to fine-tune the trade-off between mimicking the teacher model's behavior and fitting the ground truth labels, as a result optimizing the distillation process. The mathematical formula of distillation by balancing the hard loss with the soft loss is as follows:

Let Z_T and Z_S be the logits from the teacher and the student models, y be the ground truth labels, T be the temperature, and α be the weighting factor.

- Temperature scaling:

$$Z_T(T) = \frac{Z_T}{T} \quad (1)$$

$$Z_S(T) = \frac{Z_S}{T} \quad (2)$$

- Soft Loss (KL Divergence):

$$L_{Soft} = T^2 \cdot KL(\text{Softmax}(\frac{Z_T}{T}) || \log(\text{Softmax}(\frac{Z_S}{T}))) \quad (3)$$

- Hard Loss Cross Entropy:

$$L_{Hard} = \text{CrossEntropy}(Z_S, y) \quad (4)$$

- Total Loss:

$$L_{Total} = (1 - \alpha) \cdot L_{Hard} + \alpha \cdot L_{Soft} \quad (5)$$

The student model is trained using this loss function to achieve the best balance between efficiency and performance. In the next section, we demonstrate the implementation and construction of lightweight student models.

3.2.2 The Construction of Light Weight Student Models.

In our approach to constructing student models, we make several key adjustments to reduce complexity while preserving essential features. In the compressed VGG16, filter sizes are reduced to (16, 32, 64) from (64, 128, 256) for the initial, intermediate, and fully connected layers respectively. Fully connected layers are downsized from 4096 to 512 neurons. For compressing ResNet50 we use a custom residual and bottleneck block with fewer filters e.g. from (64, 128, 256, 512) to (32, 64, 128, 256), retaining 36 layers arranged in block to it(2, 2, 4 and 2) compared to the original 50. This configuration maintains the original skip connection features which is crucial for preserving gradient flow and learning deep features, while reducing the number of feature maps at each stage to lower computational cost. In student DenseNet169, we reduce the growth rate from 32 to 24 and reduce the number of layers in each dense block from (6, 12, 32, 32) to (2, 2, 2, 2). This adjustment also leads to a reduction in the number of feature map sizes, from the original (256, 512, 1280, 1664) to (64, 72, 72, 72) in the compressed version. Despite these reductions, each student model retains the core structural principles of its teacher model. This alignment helps the student models learn and generalize from the teacher's high-level representation during knowledge distillation, improving their performance despite reducing the complexity.

3.3 Implementation of Post-Training Weight Pruning Method

In the next step of our pipeline, we adopt a post-training pruning approach [22], where the weights are pruned after the model has been fully trained. Specifically, we implement a threshold-based pruning method that prunes the weights of the student models to achieve further efficient compression. This pruning technique selectively zeroes out the less important weights in the network excluding the last layer, bias, and Batch Normalization layers based on their magnitude, to maintain the structure and essential components of the model. We use a percentile-based thresholding method to adaptively prune weights which ensures robustness of the method and flexibility across different layers and models.

The mathematical formulation is following:

Let $W \in \mathbb{R}^{n \times m}$ represent the weight matrix of a particular layer. p is the pruning percentage, τ is the pruning threshold, determined by the p -th percentile of the absolute values of the weight in W . $M \in \{0, 1\}^{n \times m}$ is the binary mask for the layer. W' is the pruned weight matrix after applying the mask. $|W_{i,j}|$ refers to the absolute value of the weight at position (i, j) in the matrix.

3.3.1 Threshold Calculation:

$$\tau = \text{Percentile}(|W|, p) \quad (6)$$

This defines the pruning threshold τ , below which the weights will be pruned.

3.3.2 Binary Mask Creation:

$$M_{i,j} = \begin{cases} 1 & \text{if } |W_{i,j}| \geq \tau \\ 0 & \text{if } |W_{i,j}| < \tau \end{cases}$$

This Mask M will retain the weights greater than or equal to the threshold τ , while zeroing out the others.

3.3.3 Pruned Weight Matrix:

$$W' = W \odot M$$

The pruned weight matrix W' is obtained by the element-wise multiplication of the mask M with the weight matrix W .

The final formula is as follows:

$$W'_{i,j} = \begin{cases} W_{i,j} & \text{if } |W_{i,j}| \geq \tau \\ 0 & \text{if } |W_{i,j}| < \tau \end{cases}$$

This formula represents the core of our weight pruning algorithm

Algorithm 1 Weight Pruning with Masking

```

1: Input:
2:    $M$ : pre-trained model
3:    $p$ : Pruning percentage
4:    $D$ : Dataloader (for evaluation)
5: Initialize  $M' \leftarrow \text{deepcopy}(M)$ 
6: Initialize  $masks \leftarrow []$ 
7: for each parameter  $\theta$  in  $M'$  (excluding biases and final layers)
8:   do
9:      $W \leftarrow \theta$  {weights as a numpy array}
10:     $\tau \leftarrow \text{Percentile}(|W|, p)$  {Pruning threshold}
11:     $m \leftarrow W \geq \tau$  {Binary mask (1 if  $|W| \geq \tau$ , else 0)}
12:     $W \leftarrow W \odot m$  {Apply mask (prune weights below threshold)}
13:  end for
14: Update  $\theta$  with pruned  $W$ 
15:  $masks.append(m)$ 
16: end for
17:  $acc \leftarrow \text{Evaluate}(M', D)$  {Accuracy after pruning}
18:  $t_i \leftarrow \text{Measure\_inference\_time}(M', D)$  {Inference time after pruning}
19: Return:
20:    $(M', masks, acc, t_i)$ 
21: Output:
22:    $M'$ : Pruned model
23:    $masks$ : Masks for pruned layers
24:    $acc$ : Accuracy of pruned model
25:    $t_i$ : Inference time of pruned model

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and formalizes how the pruning percentage is used to calculate the threshold. Additionally, we create explicit pruning masks that track which weights have been pruned (set to 0). These masks provide transparency in the pruning process and can be utilized during fine-tuning to allow the reactivation of pruned weights if needed. Finally, we evaluate the pruned student model's accuracy and inference time immediately after pruning. We iterate over this process for multiple pruning rates ranging from 0% to 99%, to observe the impact on the pruned model's accuracy and inference time. By inspecting the student model's performance at different pruning levels, we gain insights into the optimal balance between zeroing out weights (weight pruning) and performance degradation. Following this feedback, we consider an optimal pruning rate to refine our student model further. The optimized, pruned model is then evaluated on the dataset to assess its classification performance and allowed for further fine-tuning if required.

4 Experiments and Results

This section presents the results and analysis of our two-step compression pipeline applied to our selected CNN models for rice leaf disease classification. This section also depicts the balance and trade-off between model compression performance across the Teacher (Base), Student (Compressed), and Pruned models.

4.1 Dataset and Training Details

The dataset used is the Paddy Leaf dataset [27], consisting of approximately 6000 samples distributed across four classes: Bacterial Blight, Blast, Brown Spot, and Tungro. The dataset is split into training (70%), validation (15%), and test (15%) sets. We analyze the outcomes of both Knowledge Distillation and Weight Pruning, focusing on model size, accuracy, inference time, and other evaluation metrics such as precision, recall, and F1 score. Hyper-parameters for the experiments include a batch size of 32, a learning rate of 0.001, and the Adam optimizer.

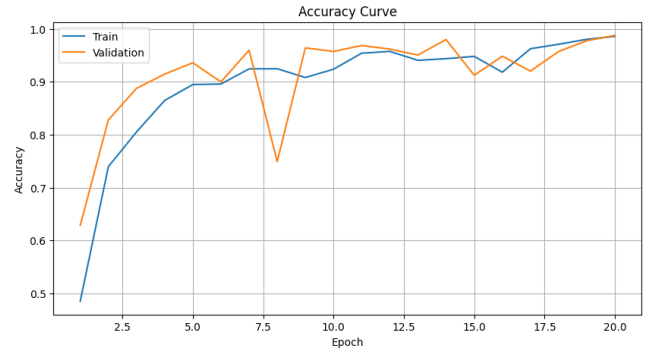
4.2 Compression using Knowledge Distillation

Our first step includes applying knowledge distillation to compress the Teacher models. As shown in Table 1, both the number of parameters and the estimated total size significantly reduced across all three models. For instance, the VGG16 Student model shows an 81% reduction in the number of parameters, while the DenseNet169 Student model achieves a remarkable 99% reduction. This reduction in parameter size can play a crucial role in model deployment on resource-constrained edge devices, without affecting the performance heavily.

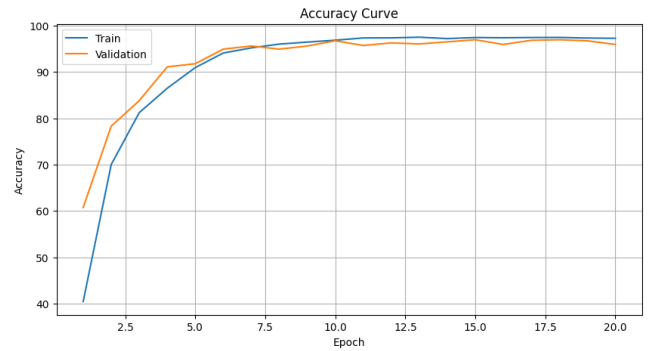
Table 1: Compression summary of teacher and student model after knowledge distillation

Variant	Model	No. of parameters	Parameter Size (MB)	Estimated Total Size MB
Teacher (Base)	VGG-16	134M	512.3	731.57
	ResNet50	23M	89.71	376.83
	DenseNet169	14M	56.60	267.80
Student (Compressed)	VGG-16	25M	98.28	144.43
	ResNet50	4M	15.53	125.12
	DenseNet169	65K	0.25	60.81

As summarized in Table 3, indicates that the models retained high accuracy despite the reduction in parameters. For example, VGG16 drops by only 2% in accuracy, while ResNet50 maintains an accuracy of 99.5%. Improvement in inference time is also noticeable, with a reduction of up to 29% in the VGG16 student model in Table 3. The minor degradation in precision, recall, and F1 score emphasizes the efficiency of knowledge distillation in maintaining classification performance. Apart from all these, we analyze the training curves for both teacher and student models to assess training stability. As illustrated in figure 2a and figure 2b the less complex student models exhibit smooth learning curves and converge similarly to the base models. This reflects the effectiveness of knowledge distillation in transferring essential knowledge to the compressed models, allowing them to perform comparably well during training and evaluation.



(a) Accuracy Curve of VGG-16 Teacher model



(b) Accuracy curve of VGG-16 Student model

Figure 2: Accuracy Curve of the CNN Model (VGG-16) after Knowledge Distillation

Table 2: Pruning summary of Compressed Student Models with optimized pruning rate.

Model (Student)	Optimal Pruning Rate (%)	No. of Pruned (Zero) parameters	No. of weighted Parameters
VGG-16	30	7.7M	18M
ResNet50	30	1.2M	2.8M
DenseNet169	15	9.6K	55.8K

4.3 Weight Pruning and Its Impact

To further reduce the student model's complexity, the second phase of our compression pipeline focuses on an adaptive threshold-based weight pruning technique. As summarized in Table 3, the models retain high accuracy despite the reduction in parameters. Improvement in inference time is also noticeable, with a reduction of up to 29% in the VGG16 student model. As shown in Figure 3a the accuracy of the models remains the same or decreases slightly for a certain amount of pruning rate, after that it decreases gradually. This gives us insight into the amount till which we can prune insignificant weights from our already compressed student models. The change in the inference time corresponding to the pruning rate is shown in the figure. 3b. By using the information from both graphs we select an optimal pruning rate for each model.

Table 3: Performance analysis of Teacher (Base), Student (Compressed), and Pruned Models

Variant	Model	Accuracy	Avg. Inference Time (sec)	Precision (Macro Avg)	Recall (Macro Avg)	F1 Score (Macro Avg)
Base (Teacher)	VGG16	99%	0.17	0.99	0.99	0.99
	ResNet50	99%	0.14	0.99	0.99	0.99
	DenseNet169	99.4%	0.16	1.00	1.00	1.00
Knowledge Distillation (Student)	VGG16	97%	0.12	0.97	0.97	0.97
	ResNet50	99.5%	0.11	1.00	1.00	1.00
	DenseNet169	98.8%	0.13	0.99	0.99	0.99
Weight Pruning	VGG16	96%	0.10	0.96	0.96	0.96
	ResNet50	99.5%	0.11	1.00	1.00	1.00
	DenseNet169	99%	0.11	0.99	0.99	0.99

Table 4: Comparison of Test Accuracies with Existing Works on Rice Disease Classification.

Author	Method	Accuracy (Overall\ Class-wise)	Disease Type
Aggarwal et al [3]	EfficientNetV2B3 with HGB Classifier on segmented data	94%	Bacterial blight, Blast, Brown spot
Ramesh et al [25]	Optimized DNN with JOA algorithm	90.57%, 95.78%, 98.9%, 94%, 92% (Class-wise)	Healthy, Bacterial blight, Blast, Brown spot, Sheath rot
Gogoi et al. [9]	3-Stage CNN using EfficientNet7 & PReLU Activation	93.99%	Healthy, Bacterial blight, Blast, Brown spot, Tungro
Sethy et al. [26]	ResNet50 with SVM	98.38%	Bacterial blight, Blast, Brown spot, Tungro
Ours	CNN (VGG16, ResNet50, DenseNet169) with Two-Step Compression Pipeline	97%, 99.5%, 99%	Bacterial blight, Blast, Brown spot, Tungro

Table 2 highlights the optimal pruning rates and the corresponding number of pruned (zero) parameters, where we can notice a significant post-training parameter reduction in the student models. The results, presented in Table 3, show that the pruned models still retain impressive performance in terms of accuracy inference time and classification, with the pruned ResNet50 model maintaining a perfect F1 score. The inference time reduction is significant as well. For example, student-VGG16 shows 41% improvement in inference time compared to its base model whereas ResNet50 and DenseNet169 show 21% and 31% improvement respectively. These results demonstrate the adaptability of our pruning strategy to maintain the balance between model compression and computational efficiency across multiple models.

From Table 4, we can see a comparison of test accuracies for various classification methods including our proposed methodology on different variants of the Paddy Leaf dataset. It includes overall accuracy for each method, with class-wise accuracy provided where available, highlighting performance across specific disease types such as Bacterial blight, Blast, Brown spot, and others.

5 Conclusion and Future Work

In conclusion, this research demonstrates the critical role of model compression techniques, specifically Weight Pruning and Knowledge Distillation, in optimizing Convolutional Neural Networks

(CNN) for Rice leaf disease classification. By significantly reducing the model size and computational demands, these techniques enable real-time analysis on resource-constrained devices, addressing key limitations of traditional CNN architectures. The optimized models not only maintain high accuracy but also provide improved inference time for instance, DenseNet169 has shown better performance among Base models, having an accuracy of 99.4%. For Student and Pruned Models, ResNet50 has outperformed other models, with an accuracy of 99.5% in both cases. In the case of inference time, the Student VGG16 shows 41% improvement compared to its base model whereas ResNet50 and DenseNet169 show 21% and 31% improvement respectively, making them highly practical for real-world agricultural applications. This research underscores the potential of integrating these techniques into agricultural disease monitoring systems, which could lead to enhanced crop productivity, minimized losses, and improved livelihoods for farmers. This study advances resource-efficient CNN models, promoting sustainable, technology-driven agriculture to combat Rice leaf diseases and enhance global food security.

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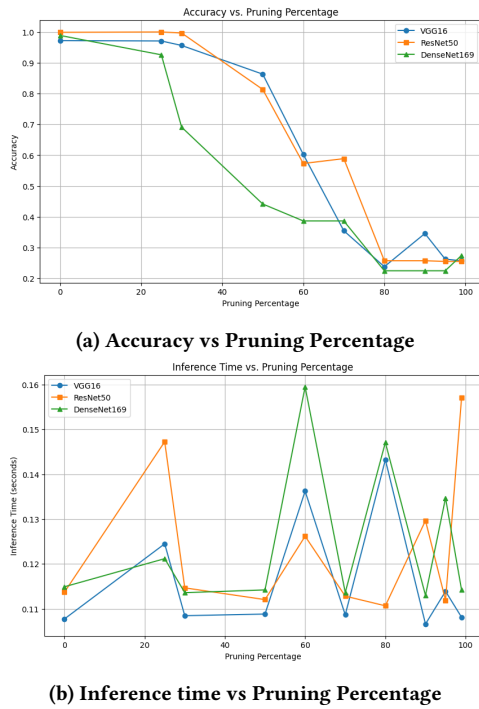


Figure 3: Accuracy and Inference Time Analysis after Weight Pruning

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